Math 110 — Week 2

Solutions

HW2 P1.

Idea. Sums can cancel, so look for the smallest cancellation.

1. The statement is false; a counterexample with $n = 1$ suffices.
2. Take $v_1 \neq 0$ and $w_1 = -v_1$. The list v_1 is independent ($\lambda v_1 = 0$ forces $\lambda = 0$ by (W3)), and so is the list w_1, since $w_1 \neq 0$.
3. $v_1 + w_1 = 0$, and a list containing 0 is dependent ($1 \cdot 0 = 0$).

HW2 P2.

Idea. (a) every axiom holds in each coordinate. (b) a finite-dimensional space cannot contain arbitrarily long independent lists. (c) finite combinations of the e_j have only finitely many nonzero entries.

1. (a) $\mathbb{F}^\infty = \mathbb{F}^{\mathbb{N}}$ with coordinatewise operations. Each vector space axiom is an equation between sequences, and it holds in every coordinate because it holds in $\mathbb{F}$. Zero vector $(0, 0, \dots)$; $-(a_j) = (-a_j)$.
2. (b) For every m, $e_1, \dots, e_m$ is independent:

$$\sum \lambda_j e_j = (\lambda_1, \dots, \lambda_m, 0, \dots) = 0$$

   forces all $\lambda_j = 0$. If $\mathbb{F}^\infty$ were spanned by a list of length n, every independent list would have length $\leq n$ (Prop. 10). Take $m = n + 1$: contradiction.
3. (c) A finite combination $\lambda_1 e_{j_1} + \cdots + \lambda_k e_{j_k}$ is 0 outside the positions $j_1, \dots, j_k$. The sequence $(1, 1, 1, \dots)$ has no zero entries, so it is not such a combination.

HW2 P3.

Idea. Start from a spanning list that lies in $U \cup W$ and thin it out; deleting vectors keeps the rest in $U \cup W$.

1. U and W are finite-dimensional (Prop. 14). Let $u_1, \dots, u_k$ be a basis of U and $w_1, \dots, w_l$ a basis of W.
2. The list $u_1, \dots, u_k, w_1, \dots, w_l$ spans $U + W = V$: every $u + w$ is a combination of the u_i plus a combination of the w_j.
3. By Prop. 9 some of its vectors can be deleted to leave a basis of V. Every remaining vector is one of the u_i or w_j, so it lies in $U \cup W$.

HW2 P4.

Idea. $\dim \mathbb{F}^2 = 2$, so a proper nonzero subspace can only have dimension 1, and two vectors form a basis exactly when they are independent.

1. (a) Let U be a subspace with $\{0\} \neq U \neq \mathbb{F}^2$. By Prop. 14, $\dim U \leq 2$, and $\dim U \neq 2$ since $U \neq \mathbb{F}^2$; $\dim U \neq 0$ since $U \neq \{0\}$. So $\dim U = 1$: U has a basis consisting of one vector $v \neq 0$, and $U = \operatorname{span}(v)$.
2. (b) ($\Rightarrow$, by the contrapositive: if one is a multiple of the other, v, w is not a basis.) If $w = cv$, then $cv - w = 0$ is a nontrivial relation, so v, w is dependent; same if $v = cw$.
3. ($\Leftarrow$) Suppose neither is a multiple of the other, and $\lambda v + \mu w = 0$. If $\mu \neq 0$ then $w = -\frac{\lambda}{\mu} v$; if $\lambda \neq 0$ then $v = -\frac{\mu}{\lambda} w$. Both are excluded, so $\lambda = \mu = 0$. Two independent vectors in a 2-dimensional space form a basis (Prop. 13).

HW2 P5.

Idea. Prop. 16 turns "$U + W$ is everything and the dimensions add up" into "the intersection is 0".

1. By Prop. 16 and $U + W = \mathbb{R}^8$,
$$\dim(U \cap W) = \dim U + \dim W - \dim(U + W) = 3 + 5 - 8 = 0.$$
2. A space of dimension 0 is $\{0\}$, so $U \cap W = \{0\}$ and $\mathbb{R}^8 = U \oplus W$ (Def. 6).

HW2 P6.

Idea. Both conditions are about the point 1, so expand p in powers of $x - 1$: then $p(1)$ and $p'(1)$ are the first two coefficients.

1. (a) The zero polynomial is in U. If $p, q \in U$: $(\lambda p + q)(1) = \lambda p(1) + q(1) = 0$ and $(\lambda p + q)'(1) = \lambda p'(1) + q'(1) = 0$.
2. (b) $1, (x-1), (x-1)^2, (x-1)^3, (x-1)^4$ is a basis of $\mathcal{P}_4(\mathbb{R})$ (Example 18), so every $p \in \mathcal{P}_4(\mathbb{R})$ can be written in exactly one way as
$$p = c_0 + c_1(x-1) + c_2(x-1)^2 + c_3(x-1)^3 + c_4(x-1)^4.$$
3. Putting $x = 1$ gives $p(1) = c_0$; differentiating once and putting $x = 1$ gives $p'(1) = c_1$. So
$$p \in U \iff c_0 = c_1 = 0 \iff p \in \operatorname{span}\big((x-1)^2, (x-1)^3, (x-1)^4\big).$$
4. These three polynomials are independent (part of a basis). So they are a basis of U and $\dim U = 3$.

R1.

Idea. Four vectors in $\mathbb{R}^4$: if they are independent they are a basis, and then both $U + W = \mathbb{R}^4$ and $U \cap W = \{0\}$ follow.

1. (a) Suppose $au_1 + bu_2 + cw_1 + dw_2 = 0$. Comparing the four coordinates,
$$a + d = 0, \qquad a + b = 0, \qquad b + c = 0, \qquad c - d = 0.$$
So $d = -a$, $b = -a$, $c = a$, and the last equation gives $a - (-a) = 2a = 0$. Hence $a = b = c = d = 0$.
2. So u_1, u_2, w_1, w_2 is an independent list of $4 = \dim \mathbb{R}^4$ vectors, a basis (Prop. 13). Hence $U + W = \mathbb{R}^4$, and $\dim U = \dim W = 2$, because u_1, u_2 and w_1, w_2 are parts of the independent list u_1, u_2, w_1, w_2, hence independent.
3. By Prop. 16, $\dim(U \cap W) = 2 + 2 - 4 = 0$, so $U \cap W = \{0\}$. With $U + W = \mathbb{R}^4$, this gives $\mathbb{R}^4 = U \oplus W$ (Def. 6).
4. (b) Solve $au_1 + bu_2 + cw_1 + dw_2 = (2, 1, 3, 0)$; comparing coordinates,
$$a + d = 2, \qquad a + b = 1, \qquad b + c = 3, \qquad c - d = 0.$$
From the last, $c = d$; then $b = 3 - d$, $a = 2 - d$, and $a + b = 5 - 2d = 1$ gives $d = 2$. So $a = 0$, $b = 1$, $c = d = 2$.
5. $u = u_2 = (0, 1, 1, 0)$ and $w = 2w_1 + 2w_2 = (2, 0, 2, 0)$. Check: $u + w = (2, 1, 3, 0)$.

R2.

Idea. Extend a basis of U to a basis of V; the added vectors span the complement.

1. Let $u_1, \ldots, u_k$ be a basis of U (Prop. 14) and extend it to a basis $u_1, \ldots, u_k, w_1, \ldots, w_l$ of V (Prop. 11). Let $W = \operatorname{span}(w_1, \ldots, w_l)$.
2. $U + W$ is a subspace containing the basis $u_1, \ldots, u_k, w_1, \ldots, w_l$ of V, so it contains every combination of it: $V \subseteq U + W$. And $U + W \subseteq V$ because $U, W \subseteq V$. So $U + W = V$.
3. If $x \in U \cap W$, then $x = \sum a_i u_i = \sum b_j w_j$, so $\sum a_i u_i - \sum b_j w_j = 0$. The combined list is independent, so all a_i, b_j are 0 and $x = 0$. So $U \cap W = \{0\}$.
4. With step 2, $V = U \oplus W$ (Def. 6).

R3.

Idea. Write the condition as one linear equation in the coefficients and solve it with free parameters.

1. For $p = a + bx + cx^2$: $p'(1) = b + 2c$ and $p(2) = a + 2b + 4c$. The condition is $a + b + 2c = 0$, i.e. $a = -b - 2c$.
2. So $p = b(x - 1) + c(x^2 - 2)$ with b, c free: $V = \operatorname{span}(x - 1,\ x^2 - 2)$. Check: for $x - 1$, $p'(1) = 1 = p(2)$; for $x^2 - 2$, $p'(1) = 2 = p(2)$.
3. They are nonzero polynomials of degrees 1 and 2, so they are independent (staircase lists are independent). So $x - 1,\ x^2 - 2$ is a basis and $\dim V = 2$.

R4.

Idea. Two spans are equal when the vectors of each list lie in the span of the other. The w's are weighted partial sums, so consecutive differences recover the v's.

1. Each $w_i = 1v_1 + 2v_2 + \cdots + iv_i$ is in $\operatorname{span}(v_1, \ldots, v_n)$. That span is a subspace, so it contains every combination of the w's: $\operatorname{span}(w_1, \ldots, w_n) \subseteq \operatorname{span}(v_1, \ldots, v_n)$.
2. $w_1 = v_1$, and for $i \geq 2$, $w_i - w_{i-1} = iv_i$. Since $i \neq 0$ in $\mathbb{R}$ and $\mathbb{C}$, $v_i = \frac{1}{i}(w_i - w_{i-1}) \in \operatorname{span}(w_1, \ldots, w_n)$. So $\operatorname{span}(v_1, \ldots, v_n) \subseteq \operatorname{span}(w_1, \ldots, w_n)$.
3. The two inclusions give equality.